



Problem 1

a) All registered voters in April 2019

b) $n = 4123$

c) Whether the registered voter approved of the President's performance or not.

d) - Proportion of all registered voters in April 2019 that approved of the President's performance.
- Categorical nominal variable.

Problem 2

$A = \{ \text{a light bulb is from supplier A} \}$

$B = \{ \text{" " " B} \}$, $IP(A) = 0.8$, $IP(D/A) = 0.001$

$C = \{ \text{" " " C} \}$, $IP(B) = 0.15$, $IP(D/B) = 0.003$

$D = \{ \text{a light bulb is defective} \}$

$$a) IP(A \cap B) = IP(D/A) \cdot IP(A)$$

$$= 0.001 \times (0.8)$$

$$= 0.0008$$

$$b) IP(D) = IP(D/A) \cdot IP(A) + IP(D/B) \cdot IP(B) + IP(D/C) \cdot IP(C)$$

$$= 0.001 \times 0.8 + 0.003 \times 0.15 + 0.005 \times 0.05$$

$$= 0.0015$$

$$c) IP(C/D') = \frac{IP(C \cap D')}{IP(D')} \\ = \frac{IP(C) - IP(C \cap D)}{1 - IP(D)}$$

$$= \frac{0.05 - (0.005 \times 0.05)}{1 - 0.0015}$$

$$= 0.0498$$

Problem 3

$$\left. \begin{array}{l} P(B) = 0.25 \\ P(C|B) = 0.15 \end{array} \right\} \text{ Given info}$$

We want $P(B'|C') = ?$

$$P(B'|C') = \frac{P(B' \cap C')}{P(C')} = \frac{P(C'|B') \cdot P(B')}{P(C')}$$

$$\cdot P(C'|B') = 1 - P(C|B')$$

$$= 1 - 0.15$$

$$= 0.85$$

$$\cdot P(C) = P(B \cap C) + P(B' \cap C)$$

$$= P(B \cap C) + P(C|B') \cdot P(B')$$

$$= 0.20 + 0.15(0.25)$$

$$= 0.2375$$

$$\Rightarrow P(B'|C') = \frac{0.85(0.25)}{1 - 0.2375} = \boxed{0.2787}$$

Problem 4

Similar to Problem 2. Only answer provided.

a) 0.027

b) 0.444

c) 0.0007

Problem 5

$C = \{ \text{Americans were affected with HCV} \}$

$I = \{ \text{" " " HIV} \}$

$$IP(C) = 0.02$$

$$IP(I|C) = 0.05$$

$$IP(C|I) = 0.12$$

$$IP(I|C) = \frac{IP(C \cap I)}{IP(C)}$$

$$\Rightarrow IP(C \cap I) = 0.001$$

$$a) IP(C|I) = \frac{IP(C \cap I)}{IP(I)} \Rightarrow IP(I) = 0.0083$$

b) No, the two viral diseases are dependent because
 $0.12 = IP(C|I) \neq IP(C) = 0.02$

Problem 6

$A = \{ \text{residents catch the usual strain of H1N1 virus} \}$

$B = \{ \text{residents catch the new strain of H1N1 virus} \}$

$$a) P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.10 + 0.05 - 0.01$$

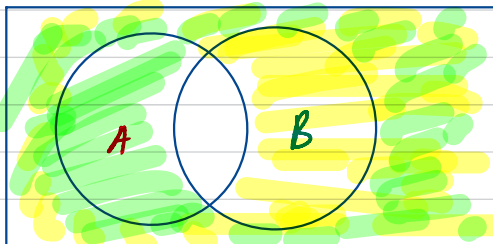
$$= 0.14$$

$$b) P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.01}{0.05} = 0.20$$

$$c) P(A'|B') = \frac{P(A' \cap B')}{P(B')}$$

$$= \frac{1 - 0.14}{1 - 0.05}$$

$$= 0.9053$$



$A' =$ 

$B' =$ 

$$P(A' \cap B') = 1 - P(A \cup B)$$

$$A' \cap B' =$$
 